

Project Milestone 2

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2024-10-25

TableOne with Standardized Mean Differences

These are the TableOne results with standardized mean differences for each row in the data set, which corresponds to one pair of back to back games. The treatment group includes only pairs of back to backs in which one of the games was missed due to load management, and the control group includes only pairs of back to backs in which both games were played. A game missed due to load management is defined as a game in which the player did not play but played in the games both before and after. This is the criteria that was chosen because teams give a variety of injury designations in these cases such as tightness or soreness rather than outright saying that it's exclusively due to rest even when it is the case that the player would have been able to play if not for the decision to manage their workload, so defining games missed due to load management this way was deemed a superior option to defining them in terms of the team's injury designation.

The covariates shown here are those that will be used to fit the propensity score model. These include:

distance_average: The average flight distance per game played in the back to back pair.

Rolling_Average_MP: The 5 game rolling average of minutes played before the back to back set.

Age: The player's age as of February 1st in the given season.

MPG: The player's minutes per game in a given season.

Positional Indicators (PG, SG, SF, PF, C): Indicators for which position the player played. This is based on the season long depth charts not individual games.

Pandemic-Influenced Season Indicators (Is_2019_20, Is_2020_21, Is_2021_22): Indicators for whether the game was in the 2019-20, 2020-21, or 2021-22 seasons. These indicators are included because each of these seasons and/or the length of the offseason leading up to them were influenced by the pandemic.

Schedule Density Indicators (Is_3rd_Game_In_4_Days, Is_4th_Game_In_5_Days, Is_5th_Game_In_7_Days): Indicators for the frequency of games as of the second game of the back to back.

When this table was initially created, it showed a couple of variables that would likely have caused data leakage if used to fit a propensity score model. The travel data was almost always empty for cases when a player didn't play a game, suggesting either that the player did not travel with the team or that the data was incorrect. In either case, this would be data leakage because any of the variables related to travel being 0 often indicated when a player sat a game, which is the outcome. Originally the intent was to include whether the player travelled to each game of the back to back and the sum of the flight distances, but now the indicators were removed and the average flight distance per game played was used instead of the sum of distances. This average should be more informative because if a player had to endure a long flight before one game of a back to back their team may have been more likely to rest them in the other game and also possibly more likely to get injured, and as expected the mean in the treatment group is slightly larger. A similar problem arose with the five game rolling average minutes played because this average includes a 0 for games not played, so it was significantly lower for the back to back pairs as a result when it was initially created including the games in the back to back set in the rolling average calculation. This was revised to avoid data leakage by using the 5 game rolling average of the games played before the back to back, so

that it would not be systematically lower for cases when a player was rested in half of the back to back. In addition, when the schedule density indicators (`Is_3rd_Game_In_4_Days`, `Is_4th_Game_In_5_Days`, `Is_5th_Game_In_7_Days`) were originally calculated, they were almost always 0 for cases when the player didn't play half of the back to back because it will never be, for example, the third game played in four nights if a player sits one of them. So these were recalculated to represent, for example, whether it would've been the third game in four nights had the player played in both games of the back to back, which is a much more meaningful covariate in the context of this study and has a much more reasonable standardized mean difference. Finally, the total games played in the season was also removed from the propensity score covariates because players who miss back to backs are playing fewer games as a result so it would be data leakage to include this in the propensity score model. However, total games played, as well as the other season-level covariates (Age, MPG, positional indicators, pandemic-influenced season indicators) will also be included in the marginal structural model that uses the propensity scores to predict the season-level injury outcomes.

		Stratified by treatment		
		0	1	SMD
##	n	17721	1316	
##	distance_average (mean (SD))	542.29 (333.14)	577.17 (538.01)	0.078
##	Rolling_Avg_MP (mean (SD))	27.95 (7.87)	25.97 (9.02)	0.234
##	Age (mean (SD))	26.17 (4.04)	28.11 (4.37)	0.462
##	MPG (mean (SD))	29.69 (4.89)	30.17 (4.71)	0.100
##	PG (mean (SD))	0.19 (0.40)	0.22 (0.42)	0.075
##	SG (mean (SD))	0.20 (0.40)	0.17 (0.38)	0.083
##	SF (mean (SD))	0.21 (0.41)	0.17 (0.38)	0.085
##	PF (mean (SD))	0.20 (0.40)	0.23 (0.42)	0.075
##	C (mean (SD))	0.20 (0.40)	0.20 (0.40)	0.011
##	Is_2019_20 (mean (SD))	0.06 (0.24)	0.07 (0.26)	0.030
##	Is_2020_21 (mean (SD))	0.09 (0.29)	0.14 (0.34)	0.142
##	Is_2021_22 (mean (SD))	0.08 (0.27)	0.14 (0.34)	0.177
##	Is_3rd_Game_in_4_Days (mean (SD))	0.68 (0.47)	0.76 (0.43)	0.168
##	Is_4th_Game_in_5_Days (mean (SD))	0.04 (0.20)	0.03 (0.18)	0.031
##	Is_5th_Game_in_7_Days (mean (SD))	0.12 (0.32)	0.16 (0.36)	0.118

ATE, Standard Error, and 95% Confidence Interval

In my project, I am comparing the results of three different groups, one in which no treatment occurred, and the other two representing the low nonzero treatment frequency and high nonzero treatment frequency, split evenly at the median of treatment frequency amongst players who had nonzero treatment frequency in the particular category. This is applied for four different categories, the early portion of the NBA season, the middle portion of the NBA season, the late portion of the NBA season, and the entire NBA season. There are two outcome variables, injury frequency and frequency of games missed, so below are the mean, standard error and 95% confidence interval for each of the three groups in each of the four categories for each of the two outcome variables. It is not practical to estimate ATE between each of these groups, but whether there is a significant difference between any two of them can be found based on whether their confidence intervals intersect.

## # A tibble: 24 x 7							
##	Outcome	Quantile	Phase	Mean	SE	CI_Lower	CI_Upper
##	<chr>	<chr>	<chr>	<dbl>	<dbl>	<dbl>	<dbl>
##	1 Injury_Frequency	Q1	Early_Quantile	0.0299	0.000886	0.0282	0.0316
##	2 Injury_Frequency	Q2	Early_Quantile	0.0384	0.00329	0.0319	0.0445
##	3 Injury_Frequency	Q3	Early_Quantile	0.0467	0.00346	0.0399	0.0532
##	4 Injury_Frequency	Q1	Mid_Quantile	0.0291	0.000923	0.0274	0.0310
##	5 Injury_Frequency	Q2	Mid_Quantile	0.0378	0.00247	0.0332	0.0429
##	6 Injury_Frequency	Q3	Mid_Quantile	0.0473	0.00321	0.0412	0.0532

```
## 7 Injury_Frequency Q1      Late_Quantile 0.0280 0.000906    0.0262    0.0298
## 8 Injury_Frequency Q2      Late_Quantile 0.0324 0.00203    0.0286    0.0364
## 9 Injury_Frequency Q3      Late_Quantile 0.0443 0.00240    0.0397    0.0491
## 10 Injury_Frequency Q1     Total_Quantile 0.0269 0.00105    0.0248    0.0288
## # i 14 more rows
```

Discussion

Data Sources

The data used in this project are from the NBA Stats website, Basketball Reference, RealGM, and Pro Sports Transactions. Specifically, the player travel data used to calculate the average flight distance was scraped from the NBA stats website using the Airball GitHub package, the data used to calculate the games played, minutes per game, rolling average minutes, age, schedule density indicators, and pandemic-influenced season indicators was scraped from Basketball Reference, the depth charts that were used to determine which players were included and the positional indicators was from RealGM, and the injury transaction data that was used to calculate the injury metrics that are the outcomes of this study were scraped from Pro Sports Transactions using the Airball GitHub package.

Inclusion/Exclusion Criteria

The inclusion and exclusion criteria that was used for the players in this study was designed to be similar to a study conducted by the NBA that was distributed to teams that investigated the relationships between frequency of game participation and injury, schedule density and injury, and cumulative NBA participation and injury. This study more specifically aims to investigate the relationship between game participation in back to back sets and injury, but similar inclusion and exclusion criteria were used to allow for loose comparison. The NBA study included 150 starting level players from each season starting in the 2012-14 season and ending in the 2022-23 season, and this study does the same. However, the starting level players considered are slightly different. This study considers players that were listed as starters on depth charts from RealGM, with slight modifications in rare cases where the depth charts were not correct due to trades.

It is also important to note that the games studied only include regular season games and do not include the bubble seeding games in July and August of 2020. These games were excluded because there was no travel for these games and there was a rest period of multiple months leading up to them due to the pandemic.

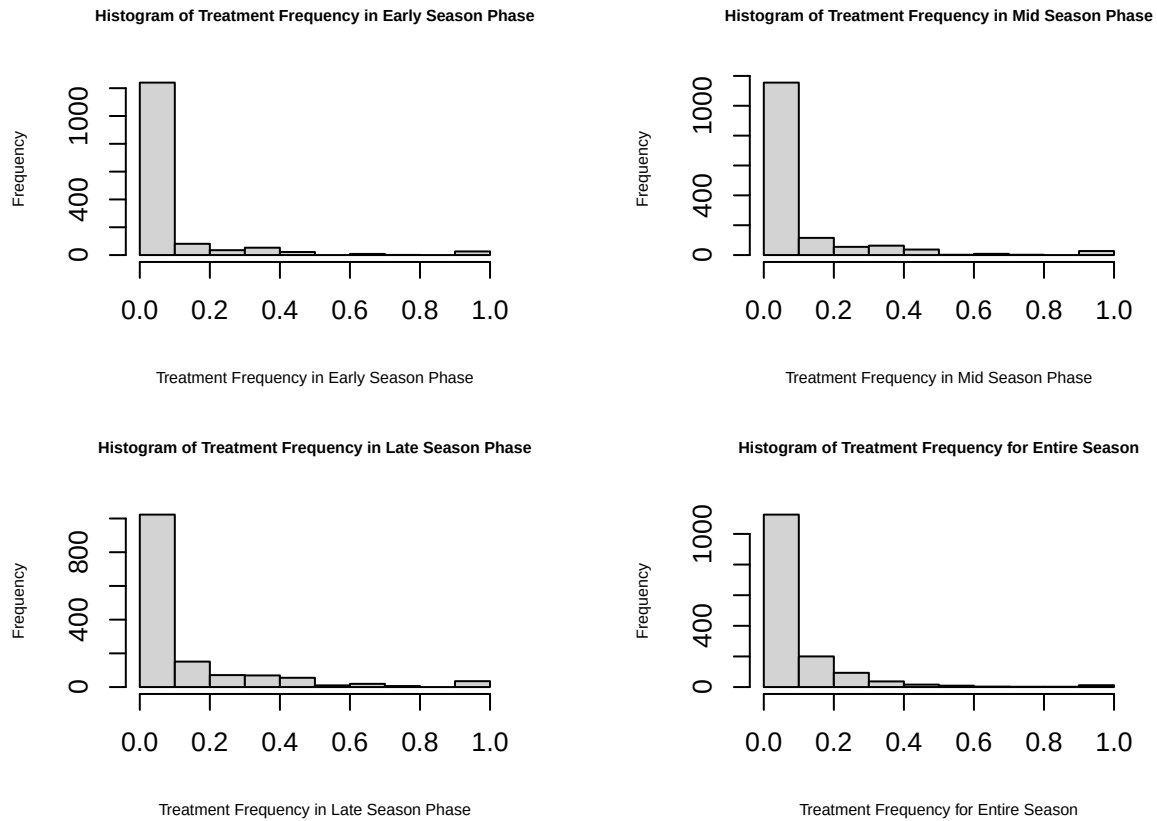
Outcomes

The outcomes are injury frequency, which is the number of games in which an injury occurred divided by the number of games played for a player in a given season, and games missed frequency, which is the number of games that a player missed due to injury divided by the number of games played by their team in the given season. It is important to note that the games missed due to injury do not include games missed due to load management. This was enforced in order to avoid data leakage in the sense that a back to back game missed, which is the treatment, would directly contribute to the games missed frequency.

Analysis

The models that will be fit for this project are a propensity score model using all of the covariates included in the TableOne and a marginal structural model (MSM) that estimates the mean of one of the outcomes in a given season phase using the treatment along with the season-level covariates and weighted using the inverse probability weights. Since the propensity scores are calculated for each game and the outcomes of the MSM are over the course of the entire season, the weights of the MSM are calculated using the sum of the products of propensity scores within the given season phase. The propensity score model will only be fit once because it is only used to predict the binary treatment of being held out of one game of a back to back. The MSM will be fit for eight different combinations of outcome and season phase. The two outcomes as defined previously are injury frequency and games missed frequency and the four season phases are early-season, mid-season, late-season, and the entire season. The first third of a team's games is the early-season phase,

the next third is the mid-season phase, and the final third is the late-season phase. These season phases are used to determine if there is significantly different outcomes based on when treatment occurs throughout the season and preserving the time-varying nature of treatment without analyzing very large numbers of possible treatment sequences. The entire season phase is included along with the early, mid, and late season phases to allow for more general analysis. For each of the two outcomes within each of the four season phases, there are three strata of treatment. The first is the case where there was no treatment in the given phase. The second and third represent a low nonzero treatment frequency in a given phase and a high nonzero treatment frequency in a given phase respectively. These were determined by splitting the frequencies to treatment at a given phase at the median amongst frequencies that are nonzero. This procedure for determining the stratification of treatment was performed in order to have meaningful distinctions between frequencies of treatment given the distribution of treatment frequency is heavily skewed right for all four season phases as shown below. The benefit of the way the treatment frequencies were stratified is that the frequency of zero represents its own category since it is by far the largest and there is still a distinction between a low nonzero frequency of treatment and a high nonzero frequency of treatment.



Missing covariate values are not an issue because the data sources are very reliable and any inconsistencies have been eliminated in the process of transforming the data. No variable selection methods will be used to fit either the propensity score model or the MSM.